PART 1

```
How many courses? (5-10)
What is the name of a course? English
What is your grade for that course?
Number grade: 100
Letter grade: A
What is the name of a course? Math
What is your grade for that course?
Number grade: 75
Letter grade: C
What is the name of a course? Spanish
What is your grade for that course?
Number grade: 65
Letter grade: D
What is the name of a course? Robotics
What is your grade for that course?
Number grade: 100
Letter grade: A
What is the name of a course? Electronics
What is your grade for that course?
Number grade: 85
Letter grade: B
Your letter grade average is: B
Your number average is: 85
```

```
/*
   Activity #6.2 Strings and Numeric Arrays
   Doniyor Nimatullo and Cole Patton
   Description: Use string arrays to create a program that calculates
   the number average of courses and outputs the letter and number average
   Started : 1/16/19
   Finished: 1/18/19
*/

String coursenames[10], lettergrades[10];
int whichselect, result, howmany, numgrades[100], sum, numaverage;

void setup() {
  Initialize();

}

void loop() {
  StartProgram();

}

void Initialize()
{
  Serial.flush();
  Serial.begin(9600);
}

void StartProgram()
{
  Serial.println("How many courses? (5-10)");
  while (Serial.available() == 0) {};
  howmany = Serial.parseInt();

  for (int i = 1; i <= howmany; i++)
  {
    Serial.print("What is the name of a course? ");
    while (Serial.available() == 0) {};
    coursenames[i] = Serial.readString();
    Serial.println(coursenames[i]); //adds names of courses to an array

    Serial.println("What is your grade for that course? ");
    while (Serial.available() == 0) {};
    numgrades[i] = Serial.parseInt(); //adds number grades to array
    Serial.print("Number grade: ");
    Serial.println(numgrades[i]);

```

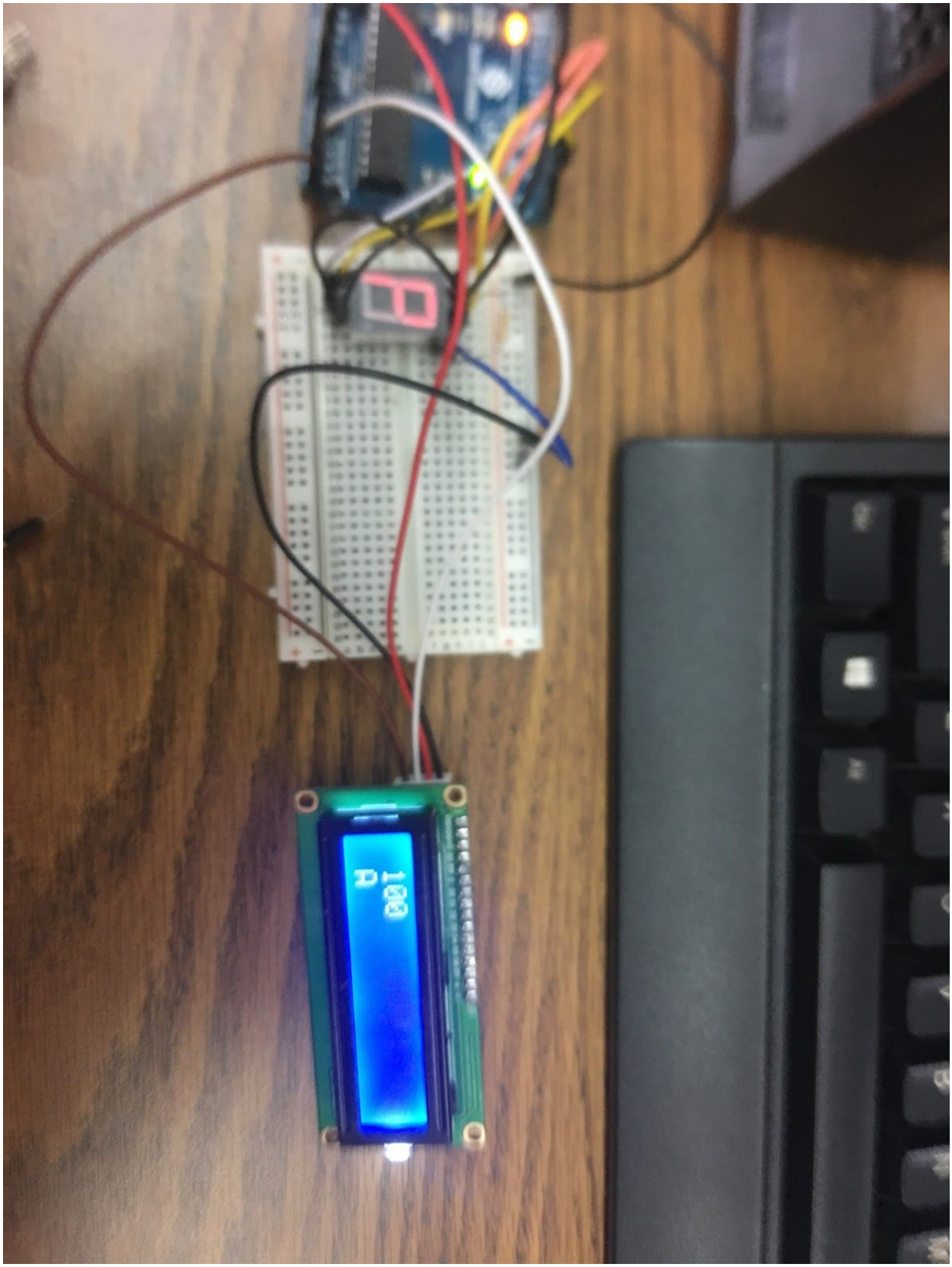
```
    switch (numgrades[i])
    {
      case 90 ... 100:
        {
          lettergrades[i] = "A";
          Serial.print("Letter grade: "); //letter grade for the class inputted not the letter grade average
          Serial.println(lettergrades[i]);
          break;
        }
      case 80 ... 89:
        {
          lettergrades[i] = "B";
          Serial.print("Letter grade: ");
          Serial.println(lettergrades[i]);
          break;
        }
      case 70 ... 79:
        {
          lettergrades[i] = "C";
          Serial.print("Letter grade: ");
          Serial.println(lettergrades[i]);
          break;
        }
      case 60 ... 69:
        {
          lettergrades[i] = "D";
          Serial.print("Letter grade: ");
          Serial.println(lettergrades[i]);
          break;
        }
      case 0 ... 59:
        {
          lettergrades[i] = "F";
          Serial.print("Letter grade: ");
          Serial.println(lettergrades[i]);
          break;
        }
      default:
        {
          break;
        }
    }
  }
  PerformCalc();
}
```

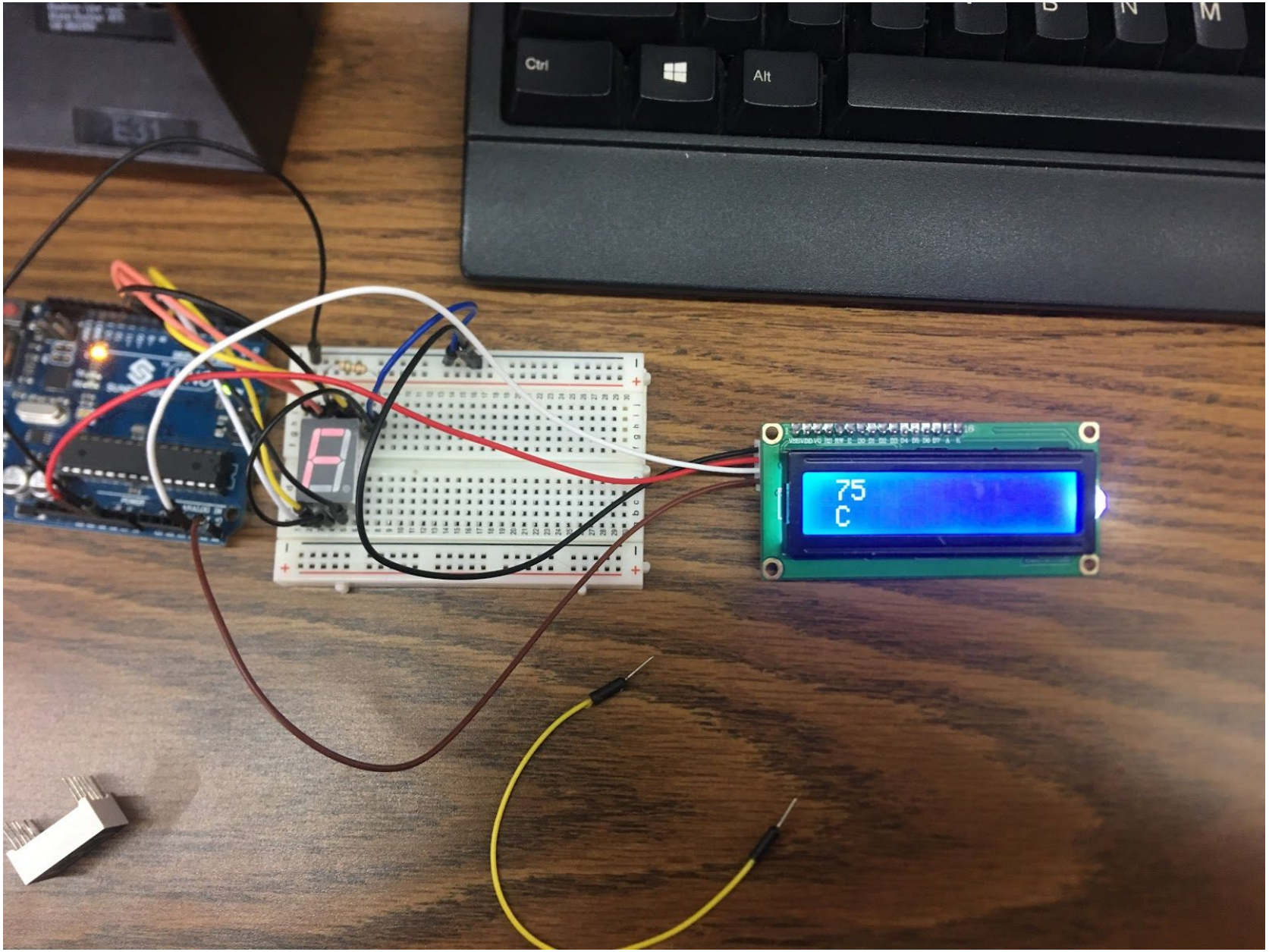
```
void PerformCalc()
{
  sum = 0;
  numaverage = 0;
  for (int i = 1; i <= howmany; i++)
  {
    sum = sum + numgrades[i]; //adds up all number grades
  }
  numaverage = sum / howmany; //finds average
  LetterAverage();
  Serial.print("Your number average is: "); //displays number average after finding it through the switch statement
  Serial.println(numaverage);
}

void LetterAverage()
{
  int i = 1;
  switch (numaverage)
  {
    case 90 ... 100:
      {
        lettergrades[i] = "A";
        Serial.print("Your letter grade average is: "); //letter grade average for all of the courses
        Serial.println(lettergrades[i]);
        break;
      }
    case 80 ... 89:
      {
        lettergrades[i] = "B";
        Serial.print("Your letter grade average is: ");
        Serial.println(lettergrades[i]);
        break;
      }
    case 70 ... 79:
      {
        lettergrades[i] = "C";
        Serial.print("Your letter grade average is: ");
        Serial.println(lettergrades[i]);
        break;
      }
    case 60 ... 69:
      {
        lettergrades[i] = "D";
        Serial.print("Your letter grade average is: ");
        Serial.println(lettergrades[i]);
        break;
      }
```

```
110 {
111   int i = 1;
112   switch (numaverage)
113   {
114     case 90 ... 100:
115       {
116         lettergrades[i] = "A";
117         Serial.print("Your letter grade average is: "); //letter grade average for all of the courses
118         Serial.println(lettergrades[i]);
119         break;
120       }
121     case 80 ... 89:
122       {
123         lettergrades[i] = "B";
124         Serial.print("Your letter grade average is: ");
125         Serial.println(lettergrades[i]);
126         break;
127       }
128     case 70 ... 79:
129       {
130         lettergrades[i] = "C";
131         Serial.print("Your letter grade average is: ");
132         Serial.println(lettergrades[i]);
133         break;
134       }
135     case 60 ... 69:
136       {
137         lettergrades[i] = "D";
138         Serial.print("Your letter grade average is: ");
139         Serial.println(lettergrades[i]);
140         break;
141       }
142     case 0 ... 59:
143       {
144         lettergrades[i] = "F";
145         Serial.print("Your letter grade average is: ");
146         Serial.println(lettergrades[i]);
147         break;
148       }
149     default:
150       {
151         break;
152       }
153   }
154 }
155
156
```

PART II


100
A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